

Confidently Wrong: Auditing and Recalibrating the Confidence of a Shipped Natural-Language-to-Shell Command Retriever

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Abstract

Natural-language shell assistants are usually built by generation. We audit a published closed-set alternative: a BM25 retriever over a fixed Windows command list that shows users a confidence and runs its choice on Enter. Its confidence is no better than a constant forecaster, and wrong answers average 86% confidence (v0.1). Under cross-validation on two benchmark versions (the second adds 59 out-of-scope and ambiguous queries), excluding 25 verbatim-copy controls, isotonic recalibration cuts the shipped confidence’s calibration error by 79% and 78%, to about a calibrated forecaster’s noise floor (outside our test family). On v0.2, a cross-validated threshold on the shipped score rejects 46 of 50 out-of-scope requests but also 20 of 134 in-scope ones; the fixed rule rejects 17 and none, a hybrid detector 34 and 11. Most out-of-scope requests are everyday ones (AI-assigned labels), and the 35 added were screened for low scores. On 4,500 unscreened external requests (pre-specified check), the threshold still beats the detector (86% vs. 78%) but, at equal cost, loses (65% vs. 77%). The hybrid’s confidence ranks its own errors better, but its accuracy gain is fragile; both systems sometimes return destructive commands for benign requests. Other analyses are exploratory; a two-annotator study is under way.

1 Introduction

Natural-language shell assistance can lower the barrier to command-line tools, but a wrong command can destroy data. Research since NL2Bash (Lin et al., 2018) has mostly generated commands: neural translation (the NLC2CMD competition, Agarwal et al., 2021), then large language model (LLM) systems (NL2SH; Westenfelder et al., 2025), most recently trained with reinforcement learning on static-analysis rewards (BashCoder-R1; Yu et al., 2026). Retrieving commands instead of generating

them is an older alternative, tested in NLC2CMD (Agarwal et al., 2021) and developed in ShellFusion (Zhang et al., 2022). Retrieval over a fixed command list cannot invent a command, and a small lexical retriever runs offline and deterministically, so queries need not leave the user’s machine and every answer is a record from a fixed list that can be inspected; the price is coverage.

We audit one such system as it is published: TermAssist, an npm package that maps a query to one of 279 Windows commands with BM25 (Robertson and Zaragoza, 2009), prints the command with a confidence percentage, refuses below 30%, and runs the pre-filled, editable command when the user presses Enter (§3). It was written and released by the author before this study, which changed none of its executable code and none of the corpus fields it reads. We evaluate the released code with the corpus in its source repository, which adds 51 records committed shortly after the release; the released corpus makes the tool less accurate and its confidence less calibrated (§3). The printed confidence is the only reliability signal a user gets, and our reproduction of the tool’s baseline outputs found it frequently confidently wrong. We therefore ask: *how reliable is the confidence of a shipped closed-set command retriever, and what do post-hoc recalibration, a tuned rejection threshold, and a hybrid lexical–dense retriever each contribute?*

Contributions. (1) An audit of a published tool’s confidence: it is no better than a constant forecaster, wrong answers average 86% confidence (v0.1), and some wrong answers are destructive commands. (2) An attribution of what fixes what, under cross-validation on two benchmark versions with verbatim-copy controls excluded: standard recalibration largely corrects the confidence level, a tuned threshold on the shipped score handles out-of-scope requests at a cost in refused in-scope queries, and a hybrid retriever (a research prototype

with one small MiniLM encoder, not part of the package) mainly improves how confidence ranks the system’s own errors. (3) The two benchmark versions, the code, and a trace from every reported number to its result file, released with the camera-ready version. The measures are established ones, applied with care for small samples: tie-aware risk-coverage areas (AURC, AUGRC), a noise floor and a no-skill reference for calibration, and out-of-scope rules tuned under one protocol and compared at equal false-rejection counts (§4, §5).

2 Related Work

Natural language to shell. NL2Bash (Lin et al., 2018) paired English descriptions with Bash commands. The NLC2CMD competition (Agarwal et al., 2021) scored each prediction by utility and flag overlap, weighted by the confidence the system submitted, so confident wrong answers were penalized; one entry used TF-IDF retrieval with a logistic-regression model that lowered confidence on likely-wrong predictions, and came within 12% of the best system. That metric folds confidence into one accuracy-like score; it does not separate calibration, the ranking of errors and out-of-scope rejection, which we measure separately. ShellFusion (Zhang et al., 2022) retrieves commands from question-answer posts, filters them with manual-page and TLDR descriptions, and ranks them against the query; DocPrompting (Zhou et al., 2023) retrieves documentation (for Bash, manual-page paragraphs) before generating code; Project CLAI (Agarwal et al., 2020) treats the command line as an environment for AI agents. NL2SH (Westenfelder et al., 2025) improves LLM translation with parsing, in-context and in-weight learning, and constrained decoding, QuoteBench (Li et al., 2026) shows that matched execution scores can hide command-path failures, and an execution-based evaluation covers generated PowerShell as well as Bash (Vo et al., 2024). We run no generative system, so our comparison with generation is conceptual.

Reliability. We draw on calibration (Guo et al., 2017; Pakdaman Naeini et al., 2015), post-hoc recalibration (Zadrozny and Elkan, 2002; Platt, 2000; Zadrozny and Elkan, 2001), selective classification and its risk-coverage metrics (Geifman and El-Yaniv, 2017; Geifman et al., 2019; Traub et al., 2024), and out-of-distribution (OOD) detection from confidence scores (Hendrycks and Gimpel,

2017). The closest task framing is intent classification with out-of-scope queries (Larson et al., 2019); the closest methods are learned calibrators for abstention under domain shift (Kamath et al., 2020) and confidence estimation for mapping language to programs (Dong et al., 2018). In text-to-SQL, selective classifiers mostly catch irrelevant questions rather than wrong queries (Somov and Tutubalina, 2025), which our out-of-scope breakdown echoes (§5). Within our targeted search (Limitations), natural-language-to-shell retrieval work has not audited the confidence a released tool shows its users, nor separated what recalibration, a tuned rejection threshold and a hybrid retriever each fix, with out-of-scope rejection weighed against its false-rejection cost; to our knowledge, that combination is what is new here, not any single method.

3 System and Benchmark

The published tool. Queries are tokenized (lowercase, all characters except ASCII letters, digits, spaces and hyphens removed, 16-word stopword list), so non-English queries are refused. Each record’s intent, category and description is scored with BM25 ($k_1 = 1.2$, $b = 0.75$), plus +15 when the query, stripped to letters and digits, is a substring of the equally stripped intent or contains it; this can fire on an accidental substring (the bare query “tar” matches “start”). Confidence is $\min(\text{round}(s/8 \times 100), 100)$ for top score s , and is 0 when $s < 2.0$. The command-line tool prints the top command with “confidence: $N\%$ ”, refuses when $N < 30$, and otherwise opens an editable prompt pre-filled with the command; a single Enter runs it through PowerShell on Windows (the default POSIX shell elsewhere). No risk level is shown. Retrieval is local; an optional query-sync feature is off by default. Our evaluation counts a query as answered when $s \geq 2.0$; no benchmark query scores between 2.0 and 2.4, so this makes the same decisions as the 30% rule. We reproduced the archived baseline outputs exactly (0/150 mismatches). Mean latency of one retrieval call, which rebuilds the index, was 3.3 ms in one run on one machine (process start-up excluded), an indicative figure.

Corpus. 431 records, of which 279 are visible on Windows (145 cross-platform, 134 Windows-specific), each a distinct intent; the tool also indexes one packaged user snippet, so its Windows index holds 280 records. The records were checked by

script rather than one by one, and a few Windows-visible records are Linux commands (e.g., `sudo reboot`). The released package has identical code but an earlier corpus of 380 records (240 visible on Windows); the 51 later records were committed shortly after the release and never published. Rerun with the released corpus, 17 of the 150 v0.1 answers change, accuracy falls from 67.3% to 58.7%, and the raw ECE of its confidence rises from 0.323 to 0.372 (v0.2: 0.293 to 0.315); 15 v0.1 items can be answered only with the later records, so we evaluate the repository corpus.

Benchmark. v0.1 has 150 hand-adjudicated queries of 9 types, labeled CORRECT (119), AMBIGUOUS (14), OOD (15; here, out-of-scope requests that no record serves) or NEEDS_CORRECTION (2), with a risk level for the gold command. All 25 *canonical* queries repeat a corpus intent verbatim and every system answers them correctly, so we treat them as positive controls and exclude them from headline figures. v0.2 contains all of v0.1 plus 35 out-of-scope and 24 ambiguous queries, so the versions share their 121 answerable queries and are not independent. The new queries were drafted and adjudicated by an AI agent under human direction, and every draft was run through both BM25 and dense retrieval: 24 of 30 ambiguous drafts were kept (6 failed the ambiguity or novelty criteria), and all 35 out-of-scope drafts were kept unchanged, their top-1 BM25 score and dense similarity having been checked to be low (maximum 7.39 and 0.31). The 15 v0.1 out-of-scope queries were instead checked by searching all corpus intents and descriptions for relevant keywords, not against retrieval scores. Seventeen of the 24 new ambiguous queries are single tool or category names (“grep”, “pip”); nine of v0.1’s 14 are too. Public natural-language-to-Bash benchmarks target Linux commands and do not map onto this Windows corpus, so we did not use them. Two items have gold commands outside the Windows corpus (TA-B145, and TA-B187 in v0.2 only), and one (TA-B149) has an acceptable command outside it; every system misses the first two and answers TA-B149 correctly, so they lower every system’s non-OOD accuracy by one query on v0.1 and two on v0.2 and change no comparison. One partially independent reviewer, shown only the query text, re-labeled a stratified sample of 14 AI-authored queries: Cohen’s $\kappa = 0.63$ (Cohen, 1960) (post hoc bootstrap 95% interval [0.39, 1.00]), be-

low our 0.7 target. The reviewer agreed with all 8 out-of-scope labels (exact 95% interval for the agreement rate [63, 100]%; 7 everyday requests and one far-from-corpus computing task, none a terminal task) and disagreed on 3 of 6 ambiguous ones. A two-annotator study with a written codebook is under way.

4 Methods

What is compared. (i) The shipped confidence, before and after post-hoc recalibration. (ii) Three out-of-scope rules: the shipped fixed rule ($s < 2.0$), a threshold on the shipped score s tuned on development folds, and the hybrid detector. (iii) Three retrievers: BM25, dense retrieval with a 384-dimensional MiniLM sentence encoder (all-MiniLM-L6-v2; Wang et al., 2020; Reimers and Gurevych, 2019), and a hybrid that min-max normalizes both score lists over each query’s candidates and ranks by $\alpha \text{BM25}_{\text{norm}} + (1 - \alpha) \text{dense}_{\text{norm}}$. The hybrid’s confidence is its fused top-1 score, which equals 1 whenever the two retrievers agree on the top command; its detector rejects when that score falls below a threshold. Appendix A gives the full specification. None of the calibrators, tuned thresholds, detectors, the risk classifier or the dense and hybrid retrievers is part of the published tool; all are offline analyses of its outputs.

Protocol. Five folds (seed 42, stratified by label; Split A). For each test fold, α (grid 0.0–1.0, maximizing non-OOD accuracy), the detector thresholds (maximizing F1) and the calibrators are chosen on the other four folds and applied once. Each threshold is chosen from the scores observed on the development folds, with out-of-scope predicted below it; ties in F1 go to the lowest such score. There is no inner split; each query’s hybrid score uses the α chosen for its own fold ($\alpha \in \{0.3, 0.5\}$ throughout). The features for the hybrid’s out-of-scope and ambiguity detectors were chosen by inspecting class means on the full data, a leak that may flatter the detector’s figures; the tuned shipped threshold uses the tool’s own score and needed no such choice. A second split groups queries by intent (Split B), which checks for tuning leakage across intents but cannot test unseen commands, since the corpus always holds the gold intent (Appendix B).

Metrics and statistics. Expected calibration error (ECE) uses 10 equal-width bins (Pak-

damani Naeini et al., 2015; Guo et al., 2017); we also report equal-mass and sweep estimates, a noise floor (the ECE of a perfectly calibrated forecaster with the same confidences and n , simulated), and Brier skill (Brier, 1950) against an out-of-fold no-skill forecaster that predicts the development base rate. Confidence often ties at its maximum (64% of the hybrid’s and 73% of the shipped values on all 150 v0.1 queries), so risk-coverage curves use expected risk under a random order within ties and are summarized by the area under them (AURC; Geifman et al., 2019) and the area under the generalized risk-coverage curve (AUGRC; Traub et al., 2024); the area under the receiver operating characteristic curve (AUROC) for separating right from wrong answers by confidence, which we call correctness AUROC, measures ranking alone. Accuracy differences use the exact McNemar test (McNemar, 1947), proportions Wilson intervals (Wilson, 1927), and effects paired percentile bootstraps (10,000 resamples, seed 42), which are not dual to the exact test. A family of four comparisons per version (accuracy vs. BM25 and vs. dense, out-of-scope rejection by the hybrid detector, hybrid ECE reduction) is Holm-corrected (Holm, 1979); it was fixed after the raw results were seen, so *all significance statements outside the external check are exploratory*. The shipped confidence’s recalibration and the tuned shipped threshold are outside this family.

External check. One analysis was specified and frozen in a tagged commit before any of its queries was scored. It applies the out-of-scope rules, unchanged, to the two test sets of CLINC150 (Larson et al., 2019), requests written for a general-purpose task assistant: 4,500 from 150 intents and 1,000 outside them. Both sets are treated as out of scope: no corpus record performs any intent’s request (checked per intent before scoring, so no intent was excluded), and the 1,000 are assumed to be. The check therefore measures rejection only, not refused in-scope queries, and is general-domain rather than terminal-specific. The tuned thresholds are v0.2’s, unchanged; all five per-fold values are applied and the median rate is reported. The primary comparison is the tuned threshold against the detector; a secondary one compares the two scores at equal v0.2 cost, each thresholded to refuse at most 11 of v0.2’s 159 in-scope queries. Intervals on the 4,500 resample intents.

5 Results

Table 1 summarizes every claim; the population throughout is non-control queries.

The shipped confidence. The published baseline answers 67.3% of all 150 v0.1 queries correctly (65.5% of the non-control in-scope ones, Table 2), yet its 49 wrong answers, none of them controls, carry a mean confidence of 86%, and 44.9% of them are shown at 100%. Its ECE (0.323 on v0.1, 0.293 on v0.2, controls excluded) is five to six times the noise floor of a perfectly calibrated forecaster at this size (0.064 and 0.050), and its Brier score is no better than predicting the base rate for every query (skill -0.23 and -0.06 , intervals including zero). A heuristic score such as $s/8$ capped at 100% is not expected to be calibrated; what the audit adds is how far off it is, and that the tool shows it to users as a percentage. Figure 1 shows the gap: 77% and 63% of non-control queries fall in the top bin, shown at a mean 99% confidence, and 70% and 73% of them are answered correctly.

Recalibration. Isotonic regression fit on development folds lowers the shipped confidence’s ECE to 0.069 on v0.1 and 0.063 on v0.2, reductions of 79% and 78%, within the noise floor’s 95th percentile (0.105 and 0.080) on this fold partition; across 20 partitions the reduction is at least 45% and 63%, but the recalibrated ECE is within that percentile on only 18 and 13 of them (Appendix B). Equal-mass bins give 0.110 and 0.070 (noise-floor 95th percentiles 0.122 and 0.092) and the bin-count sweep 0.060 and 0.047; before recalibration all three estimators give the same 0.323 and 0.293. Brier skill becomes positive (0.20 and 0.33, intervals excluding zero), so the recalibrated number carries information about correctness. Platt scaling and histogram binning, fit on the same folds, give ECEs with overlapping bootstrap intervals (v0.1: 0.087 and 0.101; v0.2: 0.045 and 0.039), but we did not test the differences between calibrators; histogram binning’s ECE is partly measured on the bins it fits, and it has the worst Brier score of the three in every case. The hybrid’s own confidence behaves similarly: raw, its Brier skill is -0.32 and -0.30 , both intervals excluding zero; after recalibration its ECE falls from 0.329 to 0.108 and from 0.368 to 0.071, above the noise floor’s 95th percentile on v0.1 (0.089) and within it on v0.2 (0.079). Its raw ECE is the same under all three estimators; after recalibration, equal-mass bins give 0.108 and

Claim	v0.1	v0.2	Status
Shipped confidence: ECE before → after isotonic	.323→.069; −79% [44, 84]	.293→.063; −78% [52, 87]	robust
Shipped confidence: Brier skill vs. no-skill, raw / recalibrated	−.23 [−.45, .01] / .20 [.04, .36]	−.06 [−.26, .15] / .33 [.19, .46]	raw: no skill; recal.: robust
Hybrid confidence: ECE before → after isotonic (Holm-family test)	.329→.108; −67% [43, 83]	.368→.071; −81% [62, 86]	robust
Out-of-scope rejected: fixed rule / tuned threshold / hybrid detector	4 / 10 / 7 of 15	17 / 46 / 34 of 50	threshold, not hybrid [†]
False rejections of in-scope: fixed / tuned / hybrid	0 / 9 / 6 of 110	0 / 20 / 11 of 134	cost of tuning
Out-of-scope rejected at equal false rejections, shipped score / hybrid feature (in-sample)	10 / 7 at 6	33 / 33 at 7; 43 / 37 at 15	descriptive [†]
Out-of-scope AUROC (pooled): shipped score minus hybrid feature	+ .102 [.006, .222]	+ .067 [.015, .127]	robust [†]
Hybrid minus shipped AURC (tie-aware)	−.115 [−.177, −.054]	−.075 [−.126, −.024]	robust
Hybrid minus shipped AUGRC (tie-aware)	−.048 [−.073, −.024]	−.030 [−.050, −.011]	robust
Hybrid minus shipped correctness AUROC	+ .099 [.008, .190]	+ .058 [−.001, .118]	weak on v0.2
Accuracy, hybrid minus BM25 (pp; Holm-adj. <i>p</i>)	+6.4 [1.8, 10.9] (.047)	+4.5 [0.7, 9.0] (.070)	fragile
Accuracy, hybrid minus dense (pp; Holm-adj. <i>p</i>)	+5.5 [0.0, 11.8] (.292)	+8.2 [2.2, 14.2] (.025)	fragile

Table 1: Claims with 95% bootstrap intervals, all on non-control queries: 110 and 134 in-scope queries plus the 15 and 50 out-of-scope ones. No rule ever rejects a control, and controls are never discordant, so the accuracy *p*-values equal the Holm table’s (Appendix B), which also holds the Holm-family ECE test. Status “robust” means the interval excludes zero on both versions; “fragile” means the result depends on a few queries. The equal-false-rejection row picks each score’s threshold on the same data (§5). [†]On the benchmark; on external requests at equal v0.2 cost the hybrid’s feature rejects more (Appendix C).

0.058 and the sweep 0.055 and 0.071. This hybrid reduction is the Holm-family comparison, tested with controls included, and it survives Holm correction on both versions (Table 5). Recalibration changes the number shown, not the answer or its ranking: correctness AUROC even drops (shipped, v0.1: 0.742 to 0.686), because each fold gets its own map. The maps are fit on benchmark queries written by the authors and an AI agent, so they hold for queries like these; a deployed calibrator would be fit on all labeled queries, and whether its map transfers to real user queries is untested.

Out-of-scope requests. The shipped fixed rule rejects 17 of v0.2’s 50 out-of-scope requests. The hybrid detector rejects 34 (two-sided exact McNemar against the fixed rule, Holm-adjusted $p = 0.00006$), at a cost of 11 false rejections among 134 in-scope queries, three of which the hybrid had answered correctly. On this benchmark, the gain comes from the threshold, not the hybrid. The shipped score itself separates out-of-scope requests better than the hybrid’s fused score (pooled AUROC 0.956 vs. 0.889 on v0.2, 0.939 vs. 0.837 on v0.1), and a threshold on the shipped score tuned

exactly like the detector rejects 46 of 50 (McNemar $p = 0.004$ against the detector) at 20 false rejections ($p = 0.093$); on v0.1 it rejects 10 of 15 and the detector 7 ($p = 0.25$), at 9 and 6 false rejections. The two tuned rules thus sit at different operating points. Compared at equal false-rejection counts, with each score’s threshold picked on the same data (an in-sample, descriptive comparison), the shipped score and the hybrid’s feature reject 23 and 21 of 50 with no false rejections, 33 and 33 at 7, and 43 and 37 at 15 (v0.1: 5 and 1 at none, 10 and 7 at 6); no control is among these false rejections. On the benchmark, the shipped score is never worse at the points we checked, and the tuned threshold’s larger count comes mostly from accepting more false rejections.

By kind of request (labels assigned by an AI assistant, unchecked), the fixed rule, tuned threshold and detector reject 14, 34 and 25 of 34 everyday non-computing requests; 3, 4 and 4 of 5 far-from-corpus computing tasks; 0, 1 and 1 of the one non-sensical request; and 0, 7 and 4 of the 10 terminal tasks the corpus does not cover. One caveat favors the shipped score: the 35 added queries were checked to have low raw BM25 scores (at most

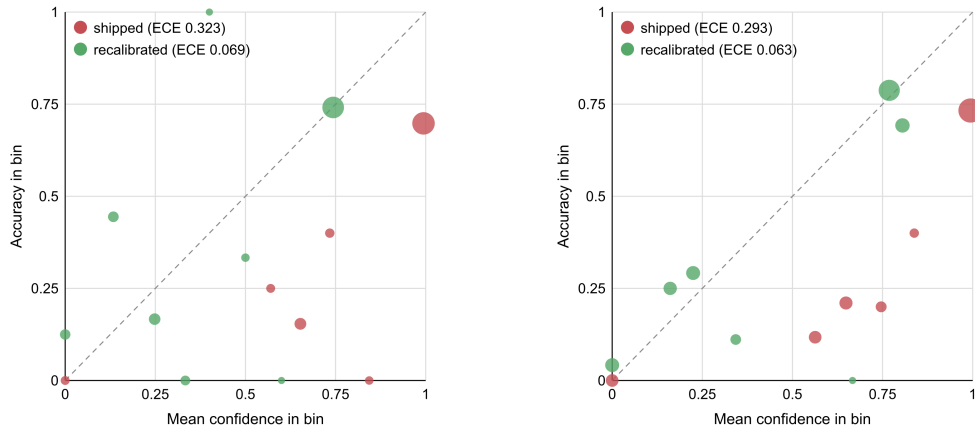


Figure 1: Reliability of the shipped confidence (red) and after isotonic recalibration fit on development folds (green), non-control queries (125 and 184), 10 equal-width bins; dot area grows with the number of queries, and the dashed line is perfect calibration. Left: v0.1. Right: v0.2.

7.39), and the tuned shipped thresholds (6.50–7.20) sit just below that maximum. The hybrid’s thresholds (0.884–0.925) are on the fused, per-query-normalized scale and cannot be compared with the 7.39 and 0.31 bounds. On the 15 original queries, which were not checked against scores, the three rules reject 4, 12 and 9 in this v0.2 run (4, 10 and 7 in the v0.1 run, Table 1); on the 35 added ones, 13, 34 and 25. The threshold’s lead over the detector therefore also appears on the unscreened queries, though 15 queries are few. (The detector’s gain over the fixed rule on the same 15 queries has $p = 0.25$ in the v0.1 run and $p = 0.0625$ in the v0.2 run, because each run tunes its own thresholds; neither is significant, and the second is not a replication.)

On the external check (Appendix C), the fixed rule rejects only 25% of the 4,500 CLINC150 requests from known intents and 19% of the 1,000 others: those that share no word, after stop-words, with the corpus, except three short greetings (e.g., “bye”) accepted through an accidental substring match (§3). At their v0.2 operating points the tuned threshold rejects 86% and the detector 78% (difference +8.3 points, interval [5.0, 11.6]; 83% and 78% on the 1,000), a smaller lead than on v0.2. At equal v0.2 cost, however, the hybrid’s feature rejects more than the shipped score (77% vs. 65%, −12.5 points [−16.2, −8.9]; 77% vs. 60% on the 1,000). On these requests, then, the benchmark’s ordering of the two scores is reversed. That fits the screening caveat above, but the populations also differ, and the check measures no false rejections.

System	v0.1	v0.2
BM25 (shipped)	72/110 (65.5)	95/134 (70.9)
Dense	73/110 (66.4)	90/134 (67.2)
Hybrid	79/110 (71.8)	101/134 (75.4)

Table 2: Non-OOD accuracy (hits/queries, %), canonical controls excluded. Every system also answers all 25 controls correctly (Appendix B).

What the hybrid adds. The hybrid’s confidence ranks its own errors better than the shipped confidence: tie-aware AURC and AUGRC are lower on both versions with intervals excluding zero (Table 1), and correctness AUROC is higher, though on v0.2 without controls its interval just includes zero. At 50% coverage, which lies inside the maximum-confidence tie block, the hybrid’s expected error on all 150 v0.1 queries is 8.3% (0–10.7% depending on how ties are ordered). Its accuracy gain is consistent in size but not established (Table 2): over BM25 it gains 6.4 points on v0.1 (exact $p = 0.0156$, Holm-adjusted 0.047) and 4.5 on v0.2 ($p = 0.070$), resting on 7 and 8 discordant queries, and one bare keyword (“system”) decides the v0.2 test, whose verdict also depends on the test used (mid-p 0.039; Appendix B); a benchmark this size has little power for gains of this size (Card et al., 2020). Its advantage over dense retrieval on v0.2 depends on three bare-keyword queries (Appendix B). Grouping folds by intent leaves accuracy unchanged and weakens calibration somewhat (Appendix B). All hybrid results are for one small encoder (MiniLM) and one fusion rule.

	v0.1		v0.2	
	Shipped	Hybrid	Shipped	Hybrid
Answered	121	125	167	184
High/critical returned	20	23	21	26
of which wrong	2	3	2	6
at max. conf.	1	1	1	1

Table 3: Returned commands the rule-based classifier rates high or critical risk, how many are wrong answers, and how many of those are shown at the system’s maximum confidence band (shipped: 100%; hybrid: fused score 1.0). As in Table 1, the population is non-control queries (125 and 184, out-of-scope requests included, since a risky command can be returned for any query); no control returns a high- or critical-risk command, so with the controls only the answered counts would change. The classifier misses at least one destructive command, so counts are lower bounds.

Risky wrong answers. The case that motivates the audit does occur (Table 3). Classifying each *returned* command with a deterministic rule-based risk classifier, both systems return a few wrong answers whose command is high or critical risk: “open the config file” returns `winget uninstall package-name -purge` from both systems, and “change the file permissions” returns an ownership change at maximum confidence. On v0.2 the hybrid also maps the bare query “system” to `sudo shutdown -h now` and the everyday request “teach me how to play guitar” to an elevated PowerShell. The full hybrid pipeline (margin plus out-of-scope rejection) blocks 3 of its 6 such answers on v0.2, including the two out-of-scope requests, but not the config, permissions or “system” answers. The classifier itself is scored against the benchmark’s own risk labels (gold commands of all non-OOD queries with a gold command: 125 in v0.1): it flags 19 of 20 high- or critical-risk commands in v0.1 and misses `git checkout -- .`, which discards uncommitted work (1/20, Wilson 95% CI [0.9, 23.6]%), v0.2 adds two misses that trace to inconsistent labels (3/22). We therefore make no safety claim. On the 15 benchmark queries whose gold commands are safe to run, executed in disposable temporary directories on the host (not an OS-level sandbox), all gold commands and 14 of 15 hybrid answers exited successfully.

6 Discussion

For a shipped closed-set assistant the three problems separate cleanly. The displayed confidence

was the weakest part, and standard recalibration on held-out labels repairs most of it on our benchmark (a result outside the corrected family); on queries like these, the number a user sees can become informative without changing the retriever. Whether a better-calibrated number changes what users run is untested. Out-of-scope rejection depended mostly on the threshold: tuned on the shipped score, it rejected more out-of-scope requests by refusing more in-scope queries, a trade-off that should be set by the relative cost of a wrong command and a refusal. Which score to threshold is not settled: the shipped score did at least as well as the hybrid’s feature at equal false-rejection counts on our benchmark (in-sample), but worse on the external requests at equal v0.2 cost. The hybrid’s value, with the one encoder we tested, lies in ranking its own errors, which matters for abstention. For users of a tool that runs a command on Enter, the risky wrong answers suggest safeguards independent of calibration. The tool runs a command only after a single Enter on a prompt pre-filled with it; beyond that, it could show the risk level and ask for confirmation (PowerShell’s `-Confirm`) or offer a dry run (`-WhatIf`) before destructive commands, or require higher confidence before returning a high-risk command. We did not evaluate any of these.

7 Conclusion

On our benchmark, the three reliability problems of a shipped closed-set retriever had three different remedies, and a better retriever was not needed for out-of-scope requests, although on external requests the hybrid’s feature rejected more at equal cost. A published natural-language command retriever shows its users a confidence that, on our benchmark, is no better than a constant forecaster. On the same benchmark, post-hoc recalibration fixes most of that. A threshold on the shipped score, tuned by cross-validation, rejects more out-of-scope requests than a hybrid detector (on v0.2, 46 vs. 34 of 50, and 12 vs. 9 on the 15 not screened by retrieval score) but refuses more in-scope queries (20 vs. 11 of 134), and most of these requests are everyday rather than terminal tasks. In an analysis frozen before scoring, on external general-domain requests, the threshold still rejects more at its operating point (86% vs. 78%), but at equal v0.2 cost the hybrid’s feature rejects more (77% vs. 65%). The hybrid, with one small encoder (MiniLM), mainly improves how confi-

dence ranks its own errors. Next steps, in order: finish the two-annotator label study and release a corrected benchmark; extend the pre-specified external check, so far general-domain, to answerable and terminal-task out-of-scope queries judged by people; build an independent safety set; and study whether calibrated confidence and risk prompts change what users run.

Limitations

- **Small, dependent, partly AI-authored benchmark.** 150 and 209 queries, the second containing the first; 59 queries drafted by an AI agent, whose labels were checked only by one partially independent reviewer ($\kappa = 0.63$, interval $[0.39, 1.00]$, $n = 14$); the two-annotator study is not yet complete. Two items have gold commands outside the corpus and one an acceptable command. Queries were written by the authors and an AI agent, not collected from users, so calibration and out-of-scope rates depend on this mix. A few accepted commands are POSIX-only and would not run in PowerShell; the shipped tool is credited with 3 such answers on v0.1 and 4 on v0.2 (e.g., `sudo reboot` for the bare query “system”), so its accuracy is slightly overstated. The tool, its corpus and the benchmark were built by the same people, so queries may follow the corpus’s phrasing more closely than users’ would; we exclude verbatim copies of corpus intents, but milder overlap may remain.
- **Exploratory statistics.** Every analysis except the external check (Appendix C) was specified after results were seen, including the Holm family, the controls exclusion, the equal-false-rejection comparison (whose thresholds are chosen in-sample) and the sensitivity subsets. Accuracy tests rest on 7–8 discordant queries against BM25 and 12–17 against dense. Headline results use one fold partition (seed 42); Appendix B repeats every fold-dependent step on 20 partitions, as point estimates.
- **Out-of-scope evidence.** Only 10 of the benchmark’s out-of-scope queries are terminal tasks outside the corpus; the kind labels are AI-assigned; the added queries were checked with raw retrieval scores, which favors the shipped score; and the hybrid detector’s fea-

tures were chosen on the full data, which may flatter its out-of-scope counts and AUROC, the ambiguity-detection results and A4/A5. The external check bears on the last two: on unscreened requests the shipped score lost its advantage at equal cost, and the detector rejected more, not less, than on v0.2. It does not bound either bias, since its population differs.

- **External check.** Its requests are general-domain, written for a task assistant, so it does not establish how the rules handle out-of-scope terminal tasks, the case that matters most for a command tool. It has no in-scope queries, so it measures rejection but not the cost in refused requests; the false-rejection counts are v0.2’s. The requests outside CLINC150’s intents are assumed out of scope, not checked one by one; the equal-cost thresholds are set in-sample on v0.2 and refuse 10 and 11 in-scope queries rather than exactly the same number; and, like the rest of the study, the check covers one tool and one corpus.
- **Scope.** One tool, evaluated with its repository corpus rather than the smaller released one (§3), one platform (Windows; results depend on the platform-filtered corpus), 279 commands, one small sentence encoder and one fusion rule, exact-match scoring against gold commands, and single-machine latencies. No generative system or user study is evaluated, since our question is the shipped retriever’s confidence, and public natural-language-to-Bash benchmarks target Linux rather than this Windows corpus.
- **Safety.** The risk classifier is scored against the benchmark’s own labels, two of which are inconsistent, and misses at least one destructive command; functional evaluation covers 15 queries and treats exit code 0 as success.
- **Literature.** Our search was targeted (natural-language-to-shell generation and retrieval, calibration, selective prediction, out-of-scope detection), last updated in October 2026; the novelty claim is limited to the combination evaluated here.

Ethical Considerations

The published tool runs the command it retrieves when the user presses Enter, and our results show

it can offer a destructive command for a benign request, sometimes at maximum confidence. We report these cases rather than a safety claim, and recommend that such tools show a risk level and require explicit confirmation for high-risk commands. This study does not change the published package, so its findings apply to current users; with the released corpus the tool is, if anything, less accurate and less calibrated (§3). No command was run on a user’s system: the functional check ran commands in disposable temporary directories on our own machine, not in an OS-level sandbox. We disclose that part of the benchmark was written by an AI agent and that our own label check fell below target. Code, benchmark files, result files and a reproduction guide will be released with the camera-ready version; they are withheld from the review version for anonymity.

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We used Claude (Anthropic), through the Claude Code tool (Claude Opus 4.8, Claude Sonnet 5 and Claude Opus 5.5), throughout this work. Under the author’s direction it drafted and revised the text of every section; wrote the analysis, test and figure-generation code and ran the analyses; drafted and adjudicated part of the benchmark queries and assigned the out-of-scope subtype labels, as disclosed in §3 and §5; simulated rounds of peer review whose suggestions led to several reported analyses (the tie-aware selective metrics, the calibration noise floor and no-skill reference, the comparison of out-of-scope rules at equal false-rejection counts, and the seed-repeat cross-validation); designed, with the author’s approval at each step, the pre-specified external check, including its equal-cost comparison; and checked the references and citations against their primary sources. No other AI tool was used for this paper’s text, code or data. The tool’s optional web dashboard, which this study does not evaluate, was built with Google Antigravity (Gemini and Claude models). The author checked the final text, its claims and its citations for accuracy and takes full responsibility for the content. No AI tool is an author of this paper.

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A System Specification

Table 4 specifies the research system; the published tool is specified in §3.

Artifacts, licenses and compute. CLINC150 is distributed under CC BY 3.0; we use its two test sets unmodified. The sentence encoder is `all-MiniLM-L6-v2` (22.7M parameters; Apache 2.0), run as its ONNX conversion through `@xenova/transformers 2.17.2` with `onnxruntime-node 1.14.0` on Node.js 24.2.0. All experiments ran on CPU on a single Windows machine; no GPU was used. Our code will be released under the MIT license and our benchmark under CC BY 4.0.

B Secondary Analyses

Holm family. Table 5 gives the Holm family, computed with the canonical controls included; the controls never enter discordant pairs, so accuracy p -values are the same without them. The calibration p is a bootstrap bound (one-sided, resolution 10^{-4}), so its Holm value is also a bound.

Bare-keyword sensitivity. Dropping single-token ambiguous queries, or all ambiguous ones (a rule stated before the subset results were computed, though the full-benchmark result was known) leaves the BM25-to-hybrid gain at +4.9 to +5.8 points, each subset 7–0 in discordant queries, but the v0.2 hybrid-over-dense advantage falls to raw $p = 0.057$ without the new bare-keyword queries and $p = 0.227$ on answerable queries only; the calibration gain is unchanged (Table 6).

Fold partitions. Every fold-dependent step (α , both thresholds and the calibrators) was repeated on 20 stratified partitions (seed 42 and seeds 1–19; non-control queries; point estimates). The ranking result holds on all 20 partitions of both versions: the hybrid’s AUROC and AUGRC are lower and its correctness AUROC higher. The tuned shipped threshold rejects more out-of-scope requests than the detector on 19 (v0.1) and 20 (v0.2) of 20 partitions (v0.2: 41–46 vs. 33–35 of 50, at 17–23 vs. 11–14 false rejections; McNemar $p < 0.05$ on 16), and the detector beats the fixed rule with $p < 0.05$ on all 20 v0.2 partitions and on none of v0.1. Isotonic recalibration cuts the shipped ECE on every partition (v0.1: 45–91%, median 82%; v0.2: 63–88%, median 78%), but the recalibrated ECE lies within the noise floor’s 95th percentile on only 18

(v0.1) and 13 (v0.2) of 20 partitions, and the hybrid’s on 9 and 11. The accuracy tests change only on the one partition per version where a fold selects $\alpha = 0.1$: hybrid over BM25 has $p = 0.016$ on 19 of 20 v0.1 partitions and $p \geq 0.07$ on all v0.2 partitions; hybrid over dense has $p = 0.013$ on 19 of 20 v0.2 partitions.

Test choice. With mid- p instead of exact McNemar tests (asymptotic tests agree), the only conclusion that changes is the v0.2 accuracy gain over BM25 (1 vs. 7 discordant queries): exact $p = 0.070$, mid- p 0.039, asymptotic 0.034, and it would survive Holm correction under mid- p (0.039). Every other comparison stays on the same side of 0.05, for example the v0.1 gain over dense (mid- p 0.092) and the v0.2 tuned threshold against the detector (mid- p 0.0024).

Split B. Grouping folds by derived intent group (171 groups on v0.2, 125 on v0.1; none split across folds) leaves hybrid accuracy unchanged and OOD AUROC nearly so, while the ECE reduction attenuates (Table 7). This table includes the controls and reports OOD AUROC as a mean over folds, so its values (0.867, 0.901) differ from the pooled, non-control values in §5. Because only 3–4 parameters are tuned and the corpus always holds the gold intent, this is a check for tuning leakage, not evidence about unseen commands.

Ambiguity and ablation. The ablation configurations are A0 (BM25), A1 (BM25 without the substring bonus), A2 (dense), A3 (hybrid), A4 (A3 plus margin rejection), A5 (A4 plus out-of-scope rejection) and A6 (A5 plus calibration, which changes confidence but no decision). Ambiguity detection is weak: F1 0.31 (v0.1) and 0.50 (v0.2), AUROC 0.78 and 0.72. Removing the substring bonus changes no top-1 answer on v0.1 and one on v0.2 (TA-B187, from one wrong command to another), so accuracy is unchanged. Adding margin rejection (A4) and out-of-scope rejection (A5) raises selective accuracy to 90.1% at 69.5% coverage on v0.1, but A5 is lower on v0.2 (82.2% at 58.9%), which we did not analyze further. Figures 2–6 show accuracy, the reliability diagrams, accuracy by query type, the risk-coverage curves and the ablation; they include the canonical controls.

Component	Definition
Dense encoder	Xenova/all-MiniLM-L6-v2; each record embedded as “Intent / Description / Category / Platform” text; cosine similarity.
Fusion	Per query, BM25 (with bonus) and cosine scores are min-max normalized over all candidates (279 corpus records and the packaged snippet, which has no dense score); $\text{score} = \alpha \text{BM25}_{\text{norm}} + (1 - \alpha) \text{dense}_{\text{norm}}$; top-1 is the argmax.
α	Grid $\{0.0, 0.1, \dots, 1.0\}$; maximizes non-OOD accuracy on the four development folds; ties go to 0.5. Selected: v0.1 0.3, 0.3, 0.3, 0.3, 0.5; v0.2 0.5 in every fold.
Hybrid confidence	Fused top-1 score (1.0 when the BM25 and dense top-1 coincide).
Margin, entropy	Top-1 minus top-2 fused score among the top 10; Shannon entropy of the top-10 scores shifted to a minimum of 0 and normalized, divided by $\log_2 10$.
Out-of-scope detector	Reject if the fused top-1 score is below a per-fold threshold maximizing F1 on development folds (v0.1 0.833–0.895; v0.2 0.884–0.925).
Tuned shipped threshold	Same protocol on the shipped top score s (v0.1 5.42–6.19; v0.2 6.50–7.20).
Ambiguity detector, A4, A5	Flag if margin is below an F1-maximizing threshold; A4 abstains on that flag, A5 on A4’s flag or the out-of-scope detector.
Calibrators	Isotonic regression (pooled adjacent violators), Platt scaling, histogram binning (10 bins); fit on development folds only.

Table 4: Specification of the research system (the hybrid is a research prototype, not part of the published tool).

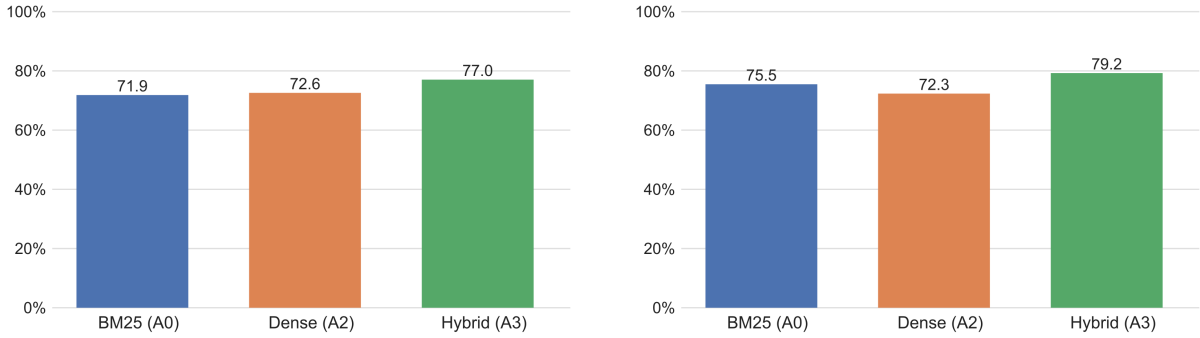


Figure 2: Accuracy on non-OOD queries (pooled; $n = 135$ and 159 , including the 25 canonical controls) for BM25, dense-only, and hybrid retrieval. Left: v0.1. Right: v0.2.

Ver.	Comparison	Raw p	Holm	Sig.
v0.1	Acc., hybrid vs. BM25	0.0156	0.0469	Yes
v0.1	Acc., hybrid vs. dense	0.1460	0.2920	No
v0.1	OOD rej., detector vs. fixed	0.25	0.2920	No
v0.1	Calib. ECE reduction	≤ 0.0001	≤ 0.0004	Yes
v0.2	Acc., hybrid vs. BM25	0.0703	0.0703	No
v0.2	Acc., hybrid vs. dense	0.0127	0.0255	Yes
v0.2	OOD rej., detector vs. fixed	0.000015	0.00006	Yes
v0.2	Calib. ECE reduction	≤ 0.0001	≤ 0.0003	Yes

Table 5: Holm-corrected family ($\alpha = 0.05$; exploratory). “OOD rej., detector vs. fixed” is the hybrid detector against the shipped fixed rule. Accuracy and out-of-scope p are two-sided exact McNemar; the calibration p is a one-sided bootstrap bound. The shipped-confidence recalibration and the tuned shipped threshold are not in this family.

C External Check on CLINC150

Table 8 gives the pre-specified results (§4). The fixed rule rejects only requests that share no word with the corpus, all of them except three short

greetings accepted through an accidental substring match; every rule rejects the rest of that group, so its share is a floor for every rule.

Ver.	Evaluated on	n	BM25→Hyb.	Dense→Hyb.	ECE red.
v0.1	all queries	135	+5.2 (0.016)	+4.4 (0.146)	80%
v0.1	w/o 9 bare-keyword	126	+5.6 (0.016)	+4.8 (0.146)	80%
v0.2	all queries	159	+3.8 (0.070)	+6.9 (0.013)	77%
v0.2	w/o 17 new bare-keyword	142	+4.9 (0.016)	+5.6 (0.057)	81%
v0.2	w/o all 26 bare-keyword	133	+5.3 (0.016)	+6.0 (0.057)	84%
v0.1/v0.2	answerable only	121	+5.8 (0.016)	+4.1 (0.227)	78/81%

Table 6: Sensitivity to bare-keyword ambiguous queries (post hoc; canonical controls included). Cells show the hybrid’s accuracy gain in points over BM25 and over dense retrieval, with the exact McNemar p (unadjusted); ECE red. is the relative ECE reduction from isotonic calibration re-fit on the subset; n counts non-OOD queries. Fusion weight and thresholds were not re-tuned.

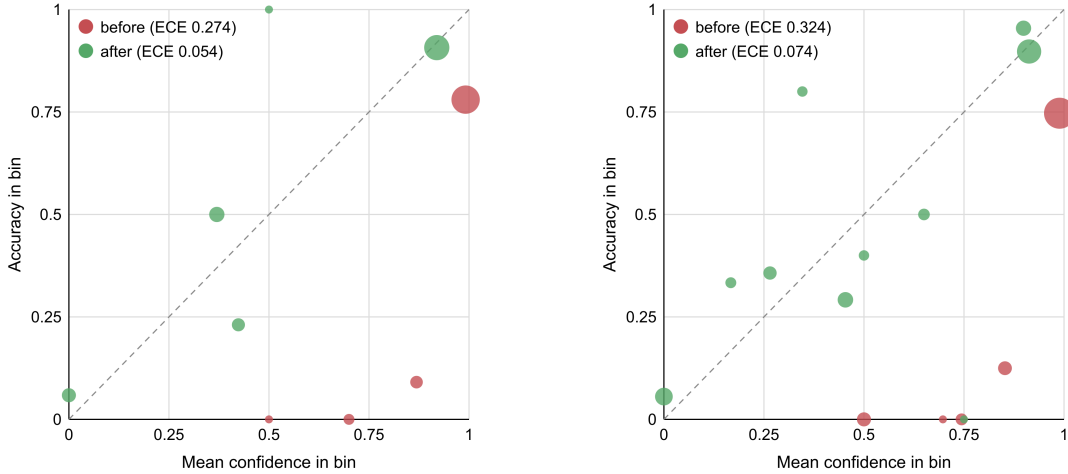


Figure 3: Reliability diagram for the hybrid system’s fused-score confidence, before vs. after isotonic calibration (10 equal-width bins; dot area grows with the number of predictions; dashed line: perfect calibration). Left: v0.1. Right: v0.2.

Ver.	Metric	Split A	Split B	Δ
v0.1	Non-OOD acc.	104/135	104/135	0
v0.1	OOD AUROC	0.867	0.849	−0.018
v0.1	Ambig. F1	0.310	0.300	−0.010
v0.1	Calib. ECE	.274→.054	.274→.065	+0.011
v0.2	Non-OOD acc.	126/159	126/159	0
v0.2	OOD AUROC	0.901	0.898	−0.003
v0.2	Ambig. F1	0.496	0.479	−0.017
v0.2	Calib. ECE	.324→.074	.324→.101	+0.027

Table 7: Split A (class-stratified) vs. Split B (grouped by intent), controls included; hybrid system. For ECE, Δ is the change in post-calibration ECE (higher is worse). AUROC values are means over folds.

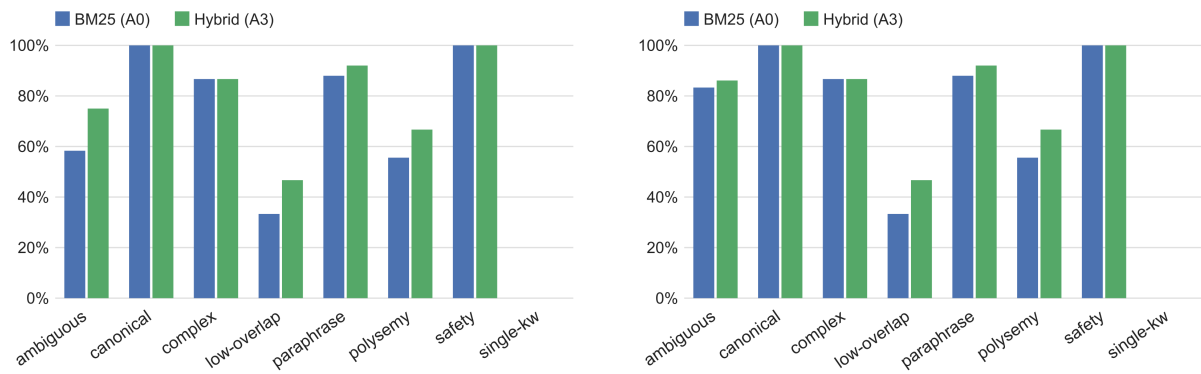


Figure 4: Accuracy by query type, BM25 (A0) vs. hybrid (A3). Single-keyword queries are 0% for both systems, so their bars are empty. Left: v0.1. Right: v0.2.

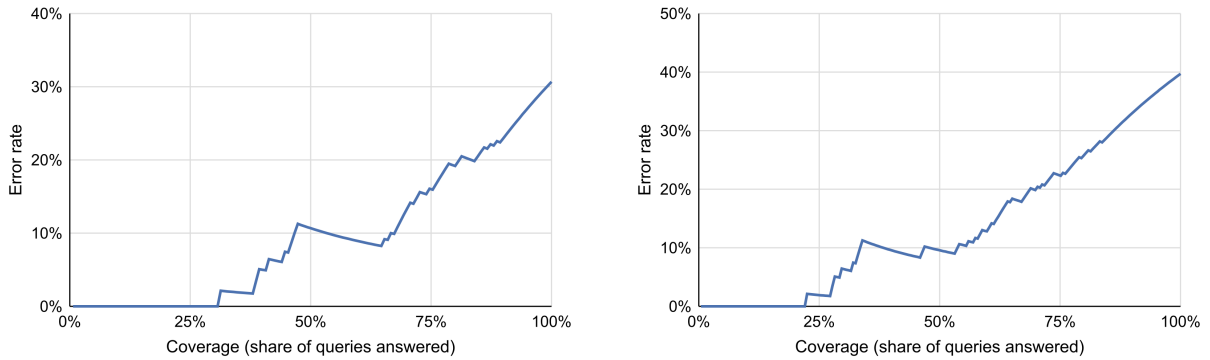


Figure 5: Risk-coverage curve for the hybrid system: queries are answered in order of decreasing confidence, and the error rate is computed over the answered share. Tied confidences are answered in benchmark order, so points inside a tie block depend on that order; tie-aware values are in the text. Left: v0.1. Right: v0.2.

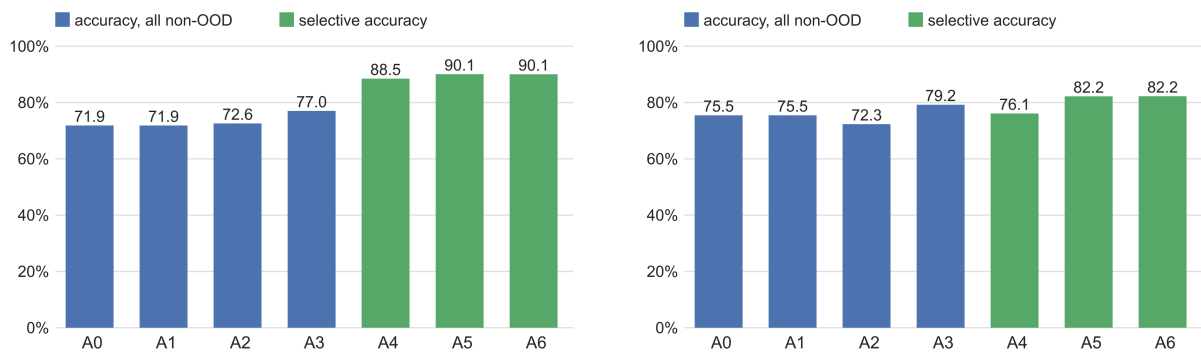


Figure 6: Ablation A0–A6. A0–A3: accuracy on all non-OOD queries (pooled). A4–A6: selective accuracy on the queries each configuration answers (mean over folds), at reduced coverage, so not directly comparable to A0–A3. Left: v0.1. Right: v0.2.

Rejected (%)	Known intents ($n = 4,500$)	Outside them ($n = 1,000$)
Fixed rule ($s < 2.0$)	25.3 [21.8, 29.0]	19.1 [16.8, 21.7]
CLI 30% rule	25.5 [22.1, 29.2]	19.8 [17.5, 22.4]
Tuned threshold (range over the five folds)	86.0 [83.0, 88.8] (86.0–92.2)	83.4 [81.0, 85.6] (83.4–89.4)
Hybrid detector (range over the five folds)	77.8 [74.1, 81.3] (69.3–78.2)	78.1 [75.4, 80.6] (69.3–78.6)
Tuned minus detector, points (primary)	+8.3 [5.0, 11.6]	+5.3 (Holm $p = .001$)
Equal v0.2 cost: shipped score / hybrid feature	64.6 / 77.1	59.7 / 77.0
difference, points	−12.5 [−16.2, −8.9]	−17.3 ($p = 6 \times 10^{-19}$)

Table 8: External check on CLINC150 (Larson et al., 2019): out-of-scope requests rejected, with 95% intervals (known intents: bootstrap over the 150 intents; outside them: Wilson; p : exact McNemar). Thresholds are v0.2’s, unchanged; the median of the five per-fold rates is shown, with their range. “Equal v0.2 cost”: each score’s threshold refuses at most 11 of v0.2’s 159 in-scope queries (10 for the shipped score, because of a tie). The protocol, including the primary comparison, was frozen before scoring, with no deviations. Removing the 13 intents a stock shell could answer leaves the primary difference at +7.5 [4.2, 10.9].